Prathyusha Bagayalla

Software Engineer

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Summary

Senior Software Engineer with **6+ years of experience** designing, developing, and maintaining **scalable, high-availability full-stack applications** across enterprise and healthcare domains. Proven expertise in building **cloud-native and microservices-based architectures** using **AWS and Microsoft Azure**. Strong background in **backend development, RESTful APIs, and event-driven systems** leveraging **Java, Python, Node.js, Kafka, and NoSQL databases**. Hands-on experience delivering responsive and user-friendly interfaces using **React, Angular, and modern web technologies**. Adept at implementing **CI/CD pipelines, containerization, and Kubernetes orchestration** to improve deployment reliability and release velocity. Demonstrated ability to optimize system performance, reduce latency, and improve uptime through **monitoring, logging, and proactive troubleshooting**. Experienced in enforcing **secure coding standards, RBAC, and compliance-focused development** in regulated environments. Strong collaborator who works closely with product managers, QA, and cross-functional teams in **Agile/Scrum environments**. Known for **end-to-end ownership**, from requirements analysis and design to production deployment and support. Passionate about writing clean, maintainable code and delivering solutions that drive measurable business impact.

Education

University of Central Missouri — Warrensburg, MO

Master of Science in Computer Science — 05/2023 - 12/2024

Skills

Programming Languages: Java, JavaScript, C/C++, C#, PL/SQL, TypeScript, Python, RDBMS

Frameworks & Libraries: Django, Phoenix, Flask, FastAPI, Express.js, NumPy, Pandas, PyTables, Matplotlib, Jinja2, Spark, Node.js, Hadoop

Web & Frontend Technologies: React, Bootstrap, HTML5, CSS3, jQuery, A JAX, GraphQL, RESTful APIs, Socket.io

Cloud & DevOps: AWS (EC2, S3, Lambda, CloudFormation, Step Functions, CloudWatch, EMR), Microsoft Azure (VMs, Azure SQL, RBAC), Docker, Kubernetes, Jenkins, GitHub Actions, AWS CDK, CI/CD Pipelines

Databases: MySQL, Oracle, SQL Server, Teradata, MongoDB, NoSQL

Testing & Automation: PyTest, Unit Test, Jest, Lettuce, Robot Framework, Snapshot Testing

Data & Analytics Tools: Apache Spark, Hadoop, Power BI, Tableau, React Dashboards, CloudWatch Metrics, Grafana

Version Control & Collaboration: Git, GitHub, Bitbucket, Jira, Agile/Scrum

Experience

Pfizer Centre One — Kansas City, KS

Software Engineer — 01/2025 - Present

- Led the design and development of **cloud-native healthcare applications**, translating business and clinical requirements into scalable technical solutions.
- Built and enhanced **full-stack applications** using Angular, JavaScript, HTML, and CSS, improving UI performance by **30%** and supporting **10K+ patient interactions**.
- Designed and optimized **backend microservices** using Java EE and Node.js, improving system reliability and reducing production defects by **28%**.
- Implemented **event-driven data pipelines** using Kafka and NoSQL databases, reducing analytics latency by **35%** and supporting real-time clinical insights.
- Owned **API design and lifecycle management**, implementing versioning, schema validation, and backward compatibility for safe deployments.
- Strengthened application security by implementing **RBAC, secure authentication, and authorization patterns**, ensuring compliance with healthcare standards.
- Improved observability by enhancing **logging, metrics, and dashboards** using CloudWatch and Grafana, reducing incident response time.
- Monitored and optimized cloud workloads across **Azure and GCP**, improving SLA visibility and performance tracking by **45%**.
- Automated build, test, and deployment workflows using **Git, Jenkins, Artifactory, and GitHub Actions**, improving release consistency by **40%**.
- Containerized and performance-tested distributed systems using **Docker, Kubernetes, and JMeter**, resolving high-load bottlenecks impacting **8K+ clinical operations**.
- Conducted root-cause analysis (RCA) for production incidents and implemented long-term fixes to improve platform stability.
- Collaborated with product managers, QA, and clinical stakeholders to define acceptance criteria and deliver compliant, production-ready features.
- Actively contributed to sprint planning, backlog grooming, and architectural discussions, influencing long-term platform scalability.
- Led end-to-end feature delivery from **requirements analysis to production deployment**, ensuring quality, scalability, and regulatory compliance.
- Designed fault-tolerant service interactions using retries, circuit breakers, and graceful degradation to improve platform resilience.
- Partnered with DevOps and SRE teams to improve **deployment strategies, rollback mechanisms, and release stability**.
- Reduced mean time to recovery (MTTR) by improving alerting, dashboards, and runbooks for critical healthcare services.
- Conducted technical reviews and contributed to **architecture decision records (ADRs)** to guide long-term platform evolution.
- Identified and implemented cost-optimization opportunities in cloud environments, improving resource utilization.

Softage Hyderabad, India

Software Engineer 05/2018 - 04/2023

- Designed and developed **enterprise-grade APIs** using ASP.NET Core and .NET Core, integrated with embedded Linux services for healthcare analytics.
- Improved data synchronization and system throughput by **40%** through optimized API design and backend processing.
- Refactored legacy monolithic applications into **modular microservices**, improving maintainability and deployment flexibility.
- Deployed and managed **containerized services** using Docker and Kubernetes, ensuring **99.9% uptime** for patient-critical systems.
- Implemented **event-driven communication** using Kafka and RabbitMQ to enable reliable inter-service messaging.
- Applied **secure coding standards, encryption, and access controls**, reducing security vulnerabilities by **30%**.
- Designed and optimized SQL schemas and queries for large healthcare datasets, enabling low-latency data access.
- Implemented caching and performance optimization techniques to improve API response times during peak traffic.
- Built responsive front-end interfaces using **React and JavaScript**, increasing user engagement by **25%**.
- Automated CI/CD pipelines using **Azure DevOps and GitHub Actions**, reducing release cycles by **35%**.
- Performed unit, integration, and regression testing using xUnit and automated test frameworks.
- Conducted performance testing and capacity planning to support high-volume clinical workloads.
- Participated in Agile ceremonies, code reviews, and cross-team collaboration to ensure delivery quality.
- Mentored junior developers and contributed to technical documentation, improving onboarding efficiency by **15%**.
- Played a key role in modernizing legacy systems by introducing **microservices, containerization, and CI/CD automation**.
- Designed reusable service templates and shared libraries, improving consistency across multiple projects.
- Supported production releases and hotfixes, ensuring minimal downtime for business-critical healthcare systems.
- Collaborated with QA teams to improve automated test coverage and reduce post-release defects.
- Assisted in cloud readiness and migration assessments, improving system reliability and disaster recovery preparedness.

Certification

- **AWS Certified Cloud Practitioner**
- **Microsoft Certified Power BI Data Analyst**
- **GraphQL & API Development Masterclass**
- **Advanced Django & Flask Development Bootcamp**